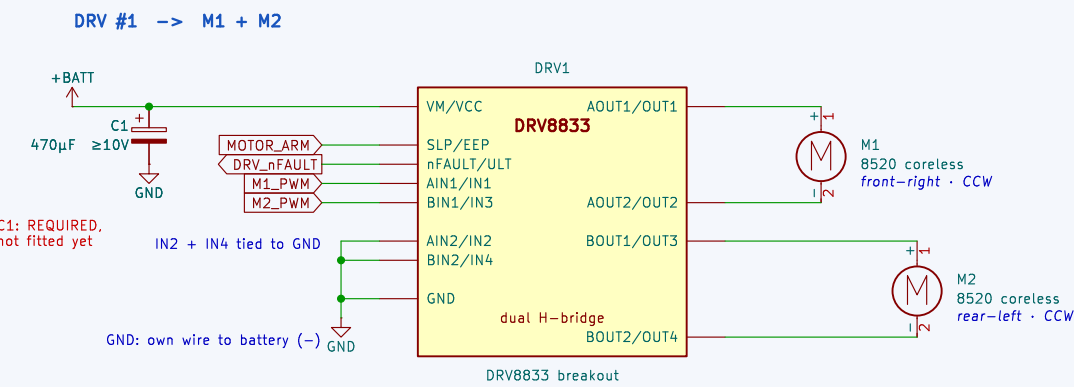


PICO DRONE – FULL WIRING

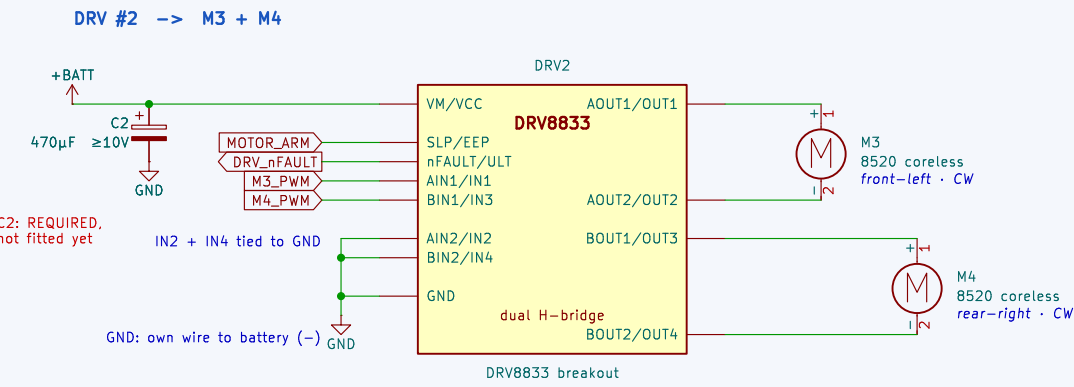
Raspberry Pi Pico 2 W · 2x DRV8833 · 4x 8520 coreless motors · GY-521 IMU · HMC5883L compass · GY-GPS6MV2 GPS (optional) · 1S LiPo

Motor wiring: STANDARD · pins from src/config.py · 2026-10-01

MOTOR DRIVERS + MOTORS



Both drivers share +BATT, GND, MOTOR_ARM (SLP) and DRV_nFAULT. Spin direction = which way round the motor leads go. Swap the two leads to reverse.

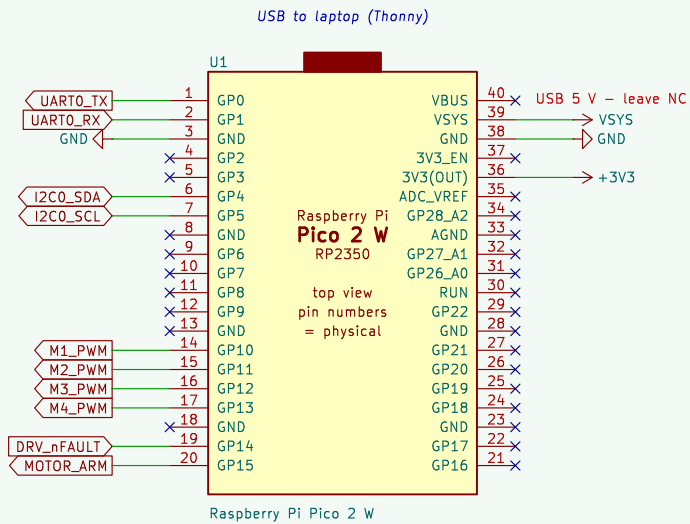


WIRING TABLE

NET	PICO 2 W PIN	CONNECTS TO
MOTOR_ARM	20 (GP15)	DRV1 SLP, DRV2 SLP
DRV_nFAULT	19 (GP14)	DRV1 nFAULT, DRV2 nFAULT (optional)
M1_PWM	14 (GP10)	DRV1 AIN1 (IN1)
M2_PWM	15 (GP11)	DRV1 BIN1 (IN3)
M3_PWM	16 (GP12)	DRV2 AIN1 (IN1)
M4_PWM	17 (GP13)	DRV2 BIN1 (IN3)
I2CO_SDA	6 (GP4)	GY-521 SDA, HMC5883L SDA
I2CO_SCL	7 (GP5)	GY-521 SCL, HMC5883L SCL
UART0_TX	1 (GP0)	GPS RX
UART0_RX	2 (GP1)	GPS TX
+3V3	36 (3V3 OUT)	GY-521 VCC, HMC5883L VCC, GPS VCC
VSYS	39 (VSYS)	D1 cathode (battery + through the SS14)
GND	38, and 3	battery (-) star point, DRV1 + DRV2 GND, sensors (GPS GND on pin 3)
+BATT	-	battery (+), DRV1 VM, DRV2 VM, D1 anode, C1 + C2 (+)
tie to GND	-	AIN2 (IN2) + BIN2 (IN4) on both drivers
motor leads	-	M1: DRV1 AOUT1/AOUT2 M2: DRV1 BOUT1/BOUT2 M3, M4: same on DRV2
(nothing)	40 (VBUS)	USB 5 V only – leave unconnected

FLIGHT CONTROLLER

Raspberry Pi Pico 2 W, top view, USB at the top. Pin numbers are PHYSICAL pins.



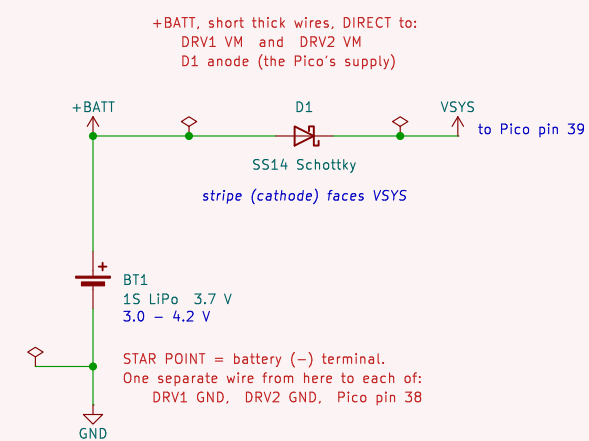
BUILD NOTES

- Battery (+) goes DIRECT to DRV1 VM and DRV2 VM. The Pico is powered only through D1 into VSYS (pin 39).
- C1, C2: 470 µF ≥10 V electrolytic, soldered right at each driver's VM/GND pins, stripe (-) to GND. REQUIRED – not fitted on the current build.
- Until C1/C2 are fitted: connect the battery BEFORE USB.
- Star ground: one wire each from battery (-) to DRV1 GND, DRV2 GND and Pico pin 38. Never daisy-chain them.
- Pin 40 (VBUS) is USB 5 V: leave it unconnected. Pin 36 (3V3 OUT) feeds the sensors only, never motors.
- MOTOR_ARM (GP15) and DRV_nFAULT (GP14) go to BOTH drivers. nFAULT uses the Pico's internal pull-up; skip it if your board has no nFAULT / ULT pin.
- Spin: M1, M2 CCW and M3, M4 CW, seen from above. Swap a motor's two leads to reverse it.
- Firmware: this sheet is STANDARD wiring, so set lift.py WIRING = "STANDARD" (it defaults to "SHARED").
- GPS is optional for v1. UART crosses over: GPS TX -> GP1 (pin 2), GPS RX <- GP0 (pin 1).
- Props OFF for every bench test except lift.py.

HOW TO READ THIS SHEET

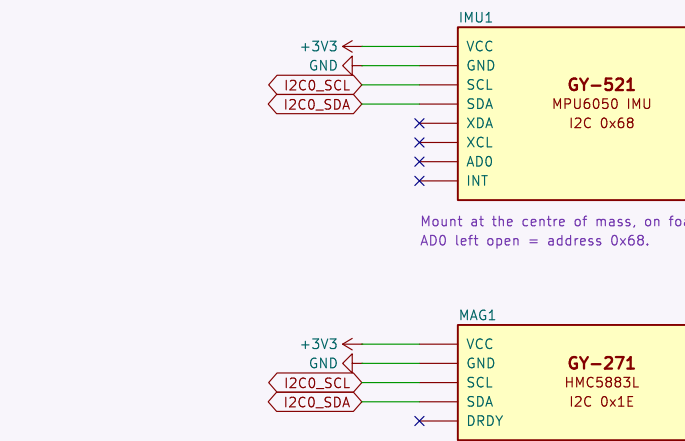
Tags with the same name (e.g. M1_PWM) are wired together. X on a pin = leave that pin unconnected. GND, +3V3, +BATT and VSYS symbols = connect to that rail. Green dot = wires joined at that point. Small diamonds (PWR_FLAG) are KiCad markers, not parts. Pico numbers 1-40 are the physical header pins.

POWER (1S LiPo)

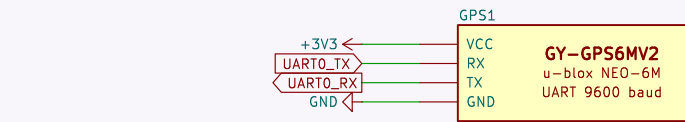


The Pico is powered only through D1 into VSYS (pin 39) – never pins 36 or 40. D1 stops USB 5 V back-feeding the LiPo when both are plugged in.

SENSORS (3.3 V from Pico pin 36)



Same I2C wires as the IMU, in parallel. Mount far from motors + battery wires. QMC5883L clone answers at 0x0D.



OPTIONAL for v1 (~14 g). TX/RX cross over. Header order varies by batch: trust the silkscreen. GND to Pico pin 3. Antenna side up.

Pico pin numbers are physical pins (USB at the top)
Pins and nets from src/config.py and firmware/README.md
Motor wiring: STANDARD
github.com/krishanth5831/pico-drone

Sheet: /
File: pico-drone.kicad_sch

Title: Pico Drone – full wiring

Size: A3 Date: 2026-10-01

KiCad E.D.A. 9.0.8

Rev: 1

Id: 1/1